

A Physics-Informed Special-Function Framework for Thermochemical Conversion Systems: Validated on Biomass Gasification

Mohsen Dirbaz

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Abstract

We present a physics-informed, special-function-parameterized reduced-order modeling framework that bridges classical special function theory with data-driven dimensional reduction for high-dimensional systems governed by conservation laws. The methodology integrates complete elliptic integrals of the first kind—computed via the arithmetic-geometric mean (AGM) algorithm—with geometric feature extraction, hierarchical sequence encoding, and neural network computational laboratories to achieve conservation closure without explicit partial differential equation (PDE) solves.

Developed and rigorously validated on biomass gasification in fluidized bed reactors, this framework demonstrates computational efficiency while maintaining physical rigor. Analysis of more than 200 experimental datasets spanning diverse fuel types (agricultural residues, wood species, energy crops) and operating conditions (temperature 600–900°C, equivalence ratio 0.2–0.5, steam-to-biomass ratio 0–1.5) yields strong predictive performance: carbon conversion efficiency ($R^2 > 0.85$), product gas yield ($R^2 > 0.82$), and individual gas species concentrations (H_2 , CO , CO_2 , CH_4 ; $R^2 > 0.78$). Critically, the framework achieves monotonic predictive relationships across full parameter ranges—eliminating local optima that plague gradient-based optimization—while requiring only ~ 5 AGM iterations per evaluation, representing a 10^5 – $10^6\times$ speedup over computational fluid dynamics.

Methodological Classification: This work occupies a unique position on the analytical-numerical spectrum. Unlike traditional numerical PDE solvers that discretize governing equations or purely data-driven surrogates that ignore physical structure, our approach *relocates* computational expense from iterative PDE integration to efficient special function evaluation. Conservation laws are enforced structurally through transfer function formulation $H(s) = k \prod (s - z_i) / \prod (s - p_j)$, where passivity, stability, and causality guarantee physical consistency by construction. Dimensional reduction via canonical correlation analysis and geometric parameterization transforms high-dimensional composition space (~ 10 variables) into a compact set of geometric invariants, which—combined with a novel “specific computational time” (sct) metric capturing computational rigidity—feed elliptic integral parameterization.

Key Innovation: The methodology demonstrates that special functions need not arise directly from governing equations to be useful. Instead, they serve as efficient “physics kernels” for parameterized reduced-order models when dimensional reduction and conservation enforcement are paramount. The framework extracts two independent universal invariants from transfer function topology: ξ (continuous log-deviation from baseline) and S (discrete structural parity).

from pole-zero configuration), both exhibiting monotonic correlations with experimental observables. Remarkably, the gain parameter k spans 10^{-47} to 10^{-60} across cases—a scale-free field numerically aligned with cosmological constant scales when normalized as $\xi = \ln(\Lambda_S/\Lambda_G)$, suggesting conservation-law systems across scales may share fundamental dimensional structures encoded in special function parameterizations.

Broader Applicability: While demonstrated for thermochemical conversion, the framework’s mathematical structure naturally extends to systems with sum-to-one constraints (mole fractions, probabilities, resource allocations), conservation laws (mass, energy, entropy), and high dimensionality requiring rapid inference. However, empirical validation is required for each new domain; structural parallels are necessary but not sufficient for generalization. This work establishes viability through rigorous thermochemical validation, providing a blueprint for extension to biochemical networks, ecological systems, or materials modeling where similar mathematical structures govern dynamics.

Keywords: Semi-Analytical Methods, Complete Elliptic Integrals, Arithmetic-Geometric Mean, Reduced-Order Modeling, Conservation Laws, Transfer Functions, Biomass Gasification, Fluidized Bed, Thermochemical Conversion, Canonical Correlation Analysis, Geometric Parameterization, High-Dimensional Data Integration